

Game Monetization Analysis

Player Payment Behavior and Purchase Activity Analysis

Author: Valeria Polishchuk

Date: May 2026

Project Objective

The objective of this project is to analyze how player characteristics and character attributes influence in-game purchase behavior, evaluate player payment activity, and identify opportunities to improve monetization strategies.

The analysis focuses on user segmentation, purchasing patterns, transaction characteristics, and product popularity.

1. Exploratory Data Analysis Results

1.1 Paying User Analysis

The dataset contains **22,214 registered players**, of which **17.7% are paying users**.

The share of paying users across character races remains relatively stable, ranging from **17.1% to 19.4%**.

However, Human players demonstrate significantly higher purchase activity in absolute numbers compared to Hobbit, Orc, and Northman players.

1.2 In-game Purchase Analysis

A total of **1,307,678 in-game purchases** were recorded with a total transaction value of **686,615,040**.

Purchase amounts vary significantly:

- Minimum purchase value: 0

- Maximum purchase value: 486,615.10
- Average purchase value: 525.69
- Median purchase value: 74.86
- Standard deviation: 2,517.35

The difference between the average and median values indicates the presence of high-value transactions affecting the average purchase amount.

After excluding zero-value transactions, the minimum purchase value becomes 0.01, which has no significant impact on overall analytical results.

1.3 Anomaly Detection

The analysis identified **907 zero-value transactions** as anomalous records.

These transactions represent only **0.07% of all purchases**, therefore they have a negligible impact on the overall analysis.

1.4 Popularity of Epic Items

The most frequently purchased epic items were:

1. **Book of Legends** — 76.87% of all purchases
2. **Bag of Holding** — 20.81% of all purchases
3. **Necklace of Wisdom** — 1.06% of all purchases

Book of Legends and Bag of Holding demonstrate the highest user adoption, with approximately **36% of paying users purchasing each item**.

2. Player Segmentation Analysis

The analysis evaluated purchasing behavior depending on character race.

Key findings:

- Human players generated the highest purchase volume (**3,921 purchases**) and demonstrated the highest purchase frequency.
- Northman players showed the highest average purchase value (**762.45**) and the highest average total spending per player (**62,520.66**).
- Orc players demonstrated the highest share of paying users among players who made purchases (**62.89%**).

These differences indicate that player segments have different purchasing patterns and monetization potential.

3. Business Recommendations

Based on the analysis, the following recommendations can be considered:

- Develop personalized offers based on player segments and purchasing behavior.
 - Focus monetization strategies on high-value customer groups, such as Northman players.
 - Analyze price sensitivity among different player segments and test targeted discounts.
 - Optimize item recommendations by promoting products with demonstrated high demand, such as Book of Legends and Bag of Holding.
-

Conclusion

The analysis demonstrated that player characteristics influence purchasing behavior and spending patterns.

Segmentation by character attributes helps identify valuable customer groups and provides opportunities for improving monetization strategies through personalized marketing campaigns and optimized pricing approaches.